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March 24, 2026

Editor-in-Chief
IEEE Transactions on Pattern Analysis and Machine Intelligence

Dear Editor-in-Chief,

We are pleased to submit our manuscript titled “**Confidence Calibration under Ambiguous Ground Truth**” for consideration for publication in *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

Summary of contributions. Standard post-hoc calibration assumes a unique ground-truth label per input and fits calibrators against majority-voted labels. We show that this practice is fundamentally flawed under label ambiguity: Temperature Scaling is theoretically biased toward temperatures that underestimate annotator uncertainty, with the true-label miscalibration gap growing monotonically with annotation entropy. To address this, we propose a family of ambiguity-aware post-hoc calibrators that optimise proper scoring rules against the full annotator distribution and require no model retraining. Our methods span three practical annotation regimes: (i) Dirichlet-Soft leverages full annotator distributions and reduces true-label ECE by 55–87% relative to Temperature Scaling across four benchmarks; (ii) Monte Carlo Temperature Scaling with a single annotation per example (MCTS $S=1$) matches full-distribution calibration within 0.6 pp ECE across all eight settings, demonstrating that pre-aggregated label distributions are unnecessary; and (iii) Label-Smooth Temperature Scaling (LS-TS) operates with voted labels alone and reduces ECE by 9–77% without any annotator data.

Relevance to TPAMI. This work addresses a fundamental gap in the evaluation and improvement of model reliability for vision and language systems operating under genuine label ambiguity. We provide both theoretical analysis and large-scale empirical validation across vision (CIFAR-10H, ISIC 2019, DermaMNIST) and language (ChaosNLI) benchmarks, covering eight architecture configurations. We believe this contribution is well-suited to the TPAMI audience, which spans pattern recognition, computer vision, and machine learning.

Originality. This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with the submission to TPAMI. The work does not raise any ethical concerns; all datasets used are publicly available or derived from clinician-reported statistics in the published literature.

We suggest the manuscript be classified as a **Regular Paper**.

We look forward to receiving the reviewers’ feedback and thank you for your time and consideration.

Sincerely,

Linwei Tao
on behalf of all authors